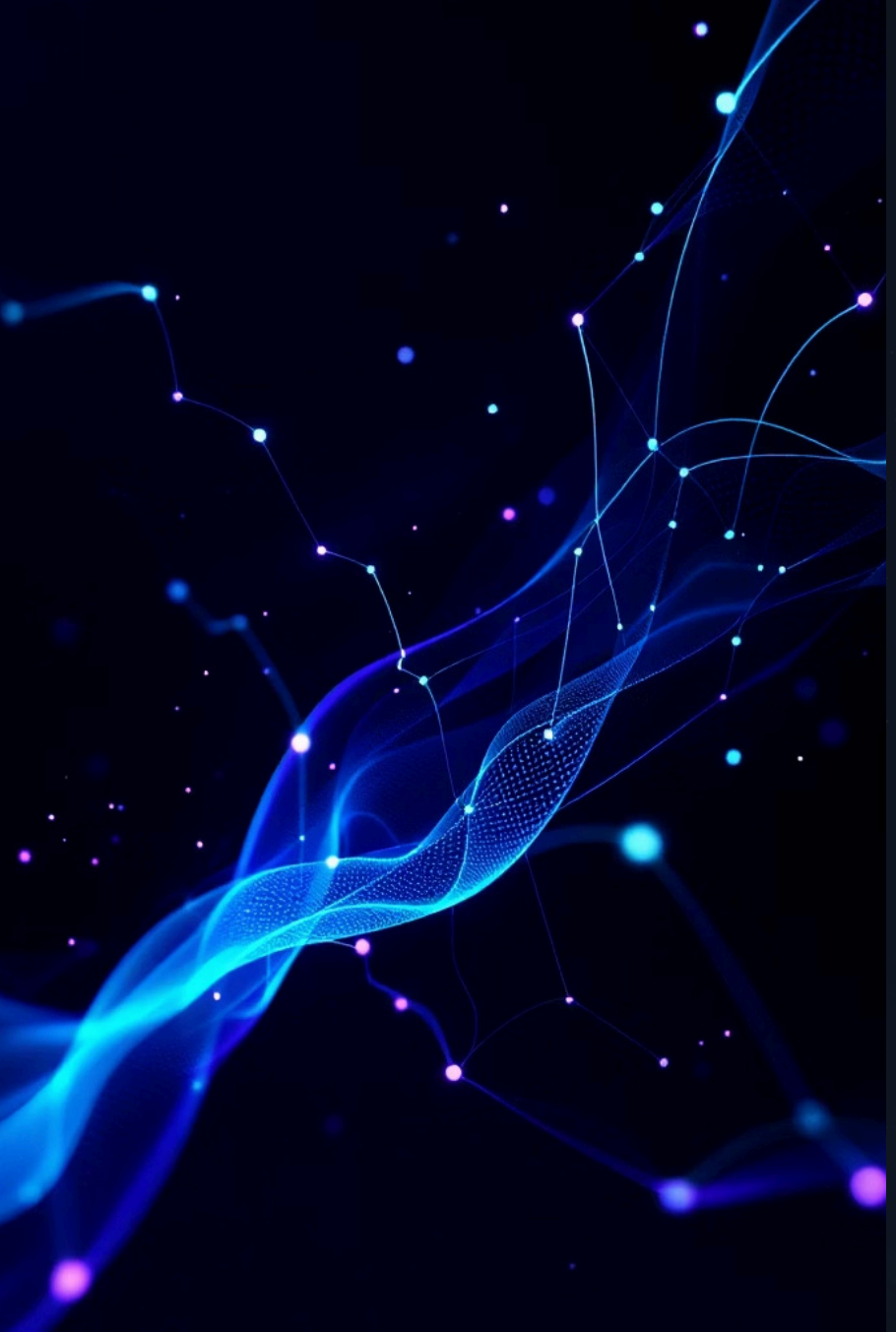

Context-Aware Smart Attendance System

Using Geo-Fencing, Dynamic Session Authentication & Motion-Based Validation

Final-year BTech CSE project — concise technical briefing for examiners and student reviewers

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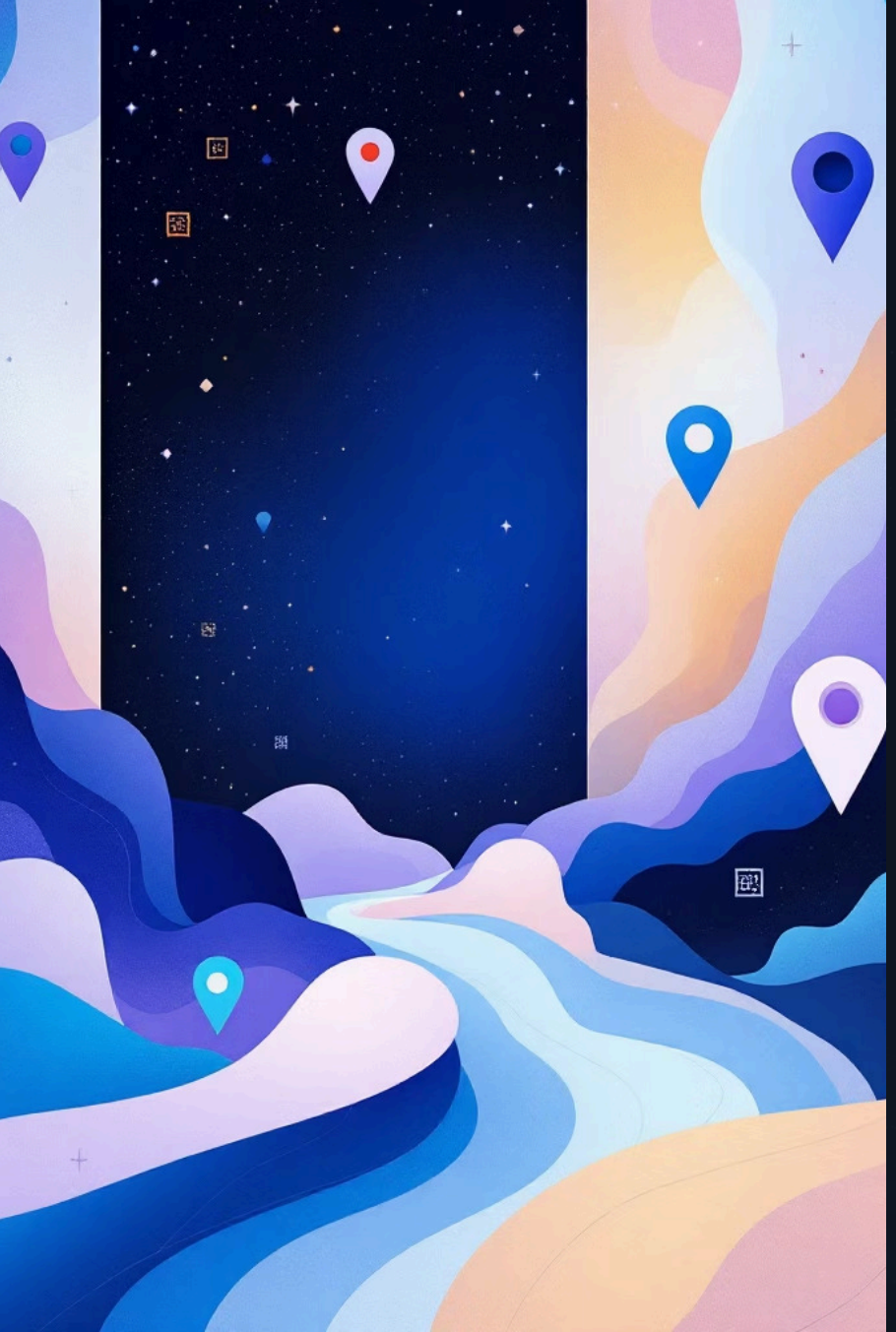

1. Problem Statement

Attendance Integrity

Proxy marking and insincere roll-calls undermine record accuracy.

Location Ambiguity

Manual checks can't verify attendee proximity to classroom precisely.

Session Spoofing

Static QR codes or OTPs can be copied or replayed by attackers.

Scalability & Privacy

Need solutions that scale across large cohorts while preserving PII.

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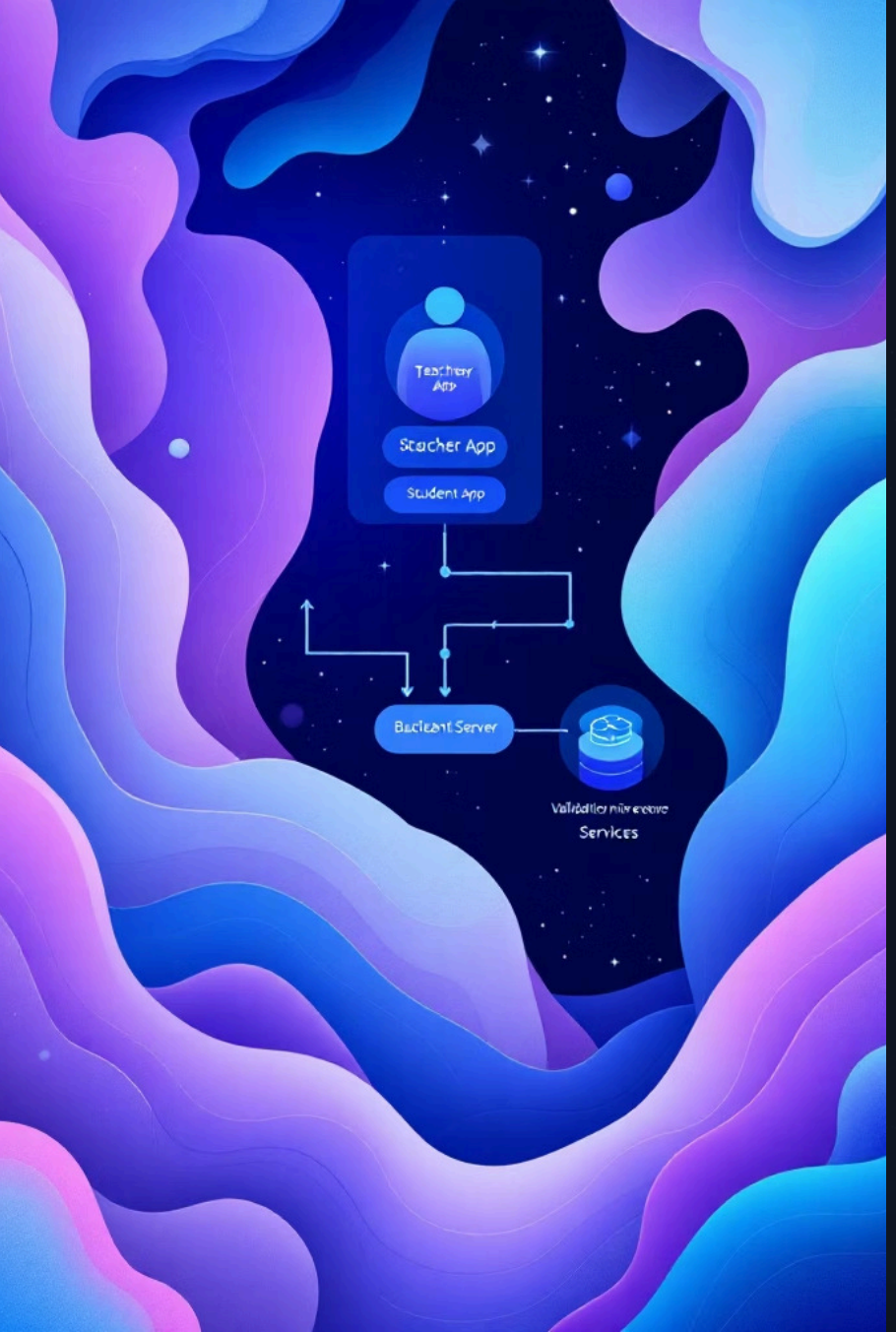

2. Proposed Solution — Overview

Multi-layered, context-aware attendance platform combining:

- Geo-fencing to assert physical presence
- Dynamic per-session QR authentication
- Motion-based sensor validation (accelerometer + gyroscope)
- Server-side multi-factor verification with audit logging

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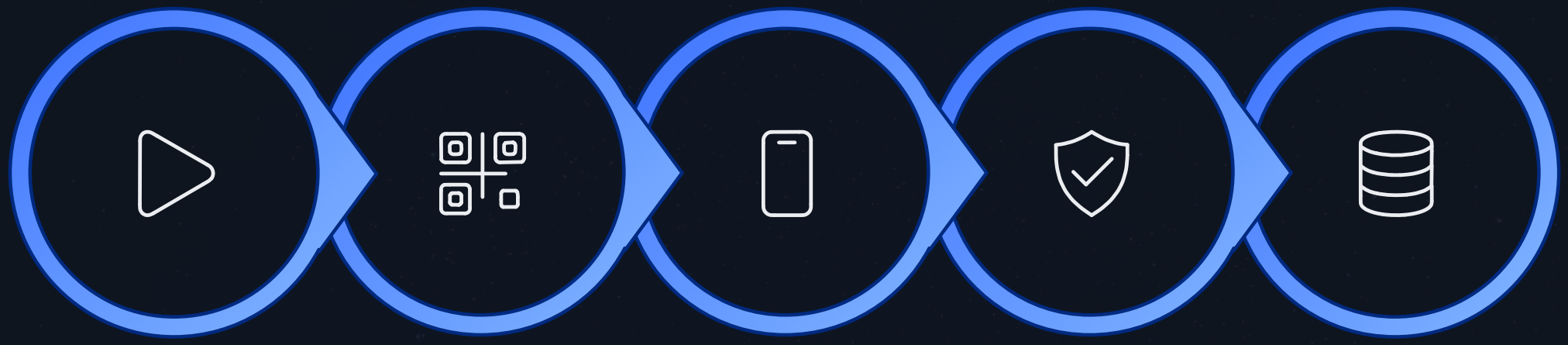

Start Session → **Generate QR & Geo** → **Student Check-In** → **Multi-Layer Validate** → **Store & Audit**

Each step enforces a distinct context: temporal, spatial, biometric-like motion, and cryptographic session proof. Backend applies sequential validations before persistence.

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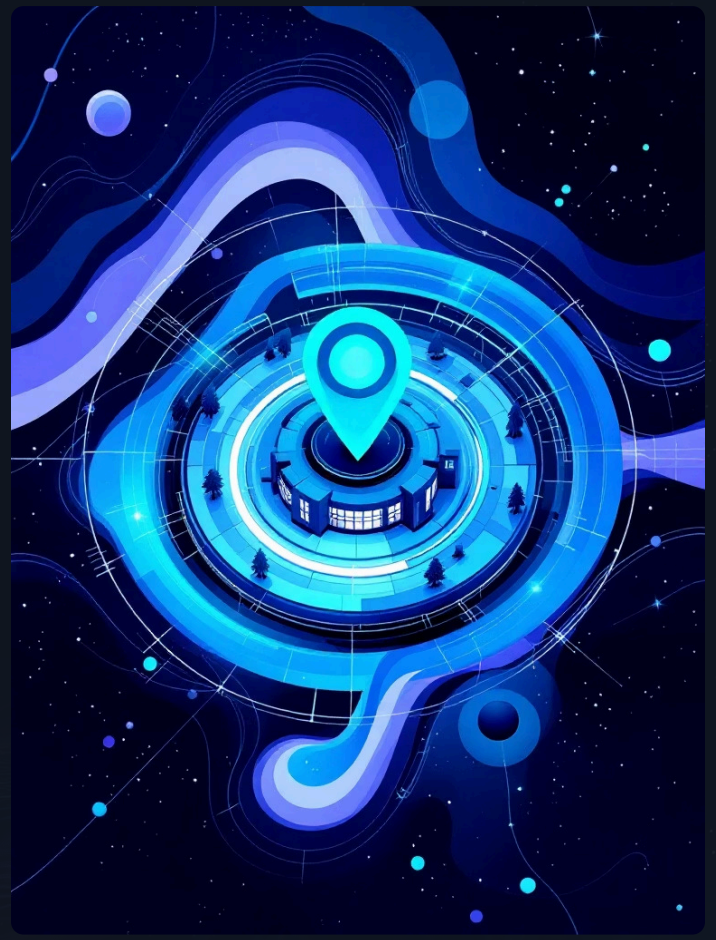

4. Geo-Fencing Layer

Defines a secure perimeter per classroom or venue. Key aspects:

- Radius configured per venue (GPS accuracy & error margin)
- Server-verified geolocation snapshot (lat, long, timestamp)
- Adaptive radius: expands slightly for urban GPS noise, shrinks indoors when Wi-Fi fingerprinting available
- Rejects check-ins outside permitted window or boundary

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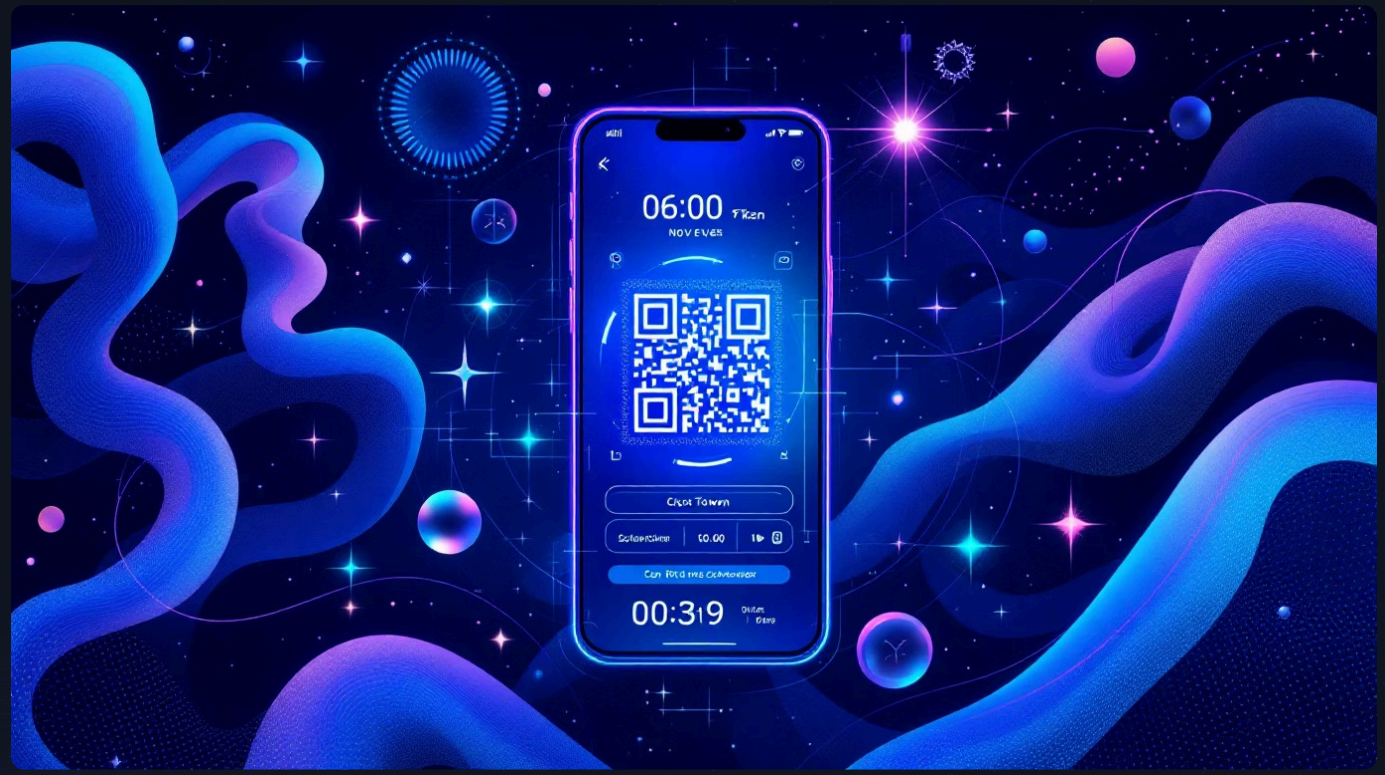

5. Dynamic QR Authentication

Per-session QR codes change frequently and embed signed metadata:

- Contents: sessionID, expiry, nonce, truncated HMAC (server-signed)
- Short TTL (e.g., 30–60 seconds) to prevent replay
- Rotation driven by server; teacher app displays live QR
- Student app verifies QR structure locally, then submits scan to server for signature check

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6. Motion Sensor Validation

Novel layer to detect genuine user interaction and device context:

Accelerometer Snapshot

Short vector sample during scan — verifies characteristic human handling motion.

Gyroscope Pattern

Angular velocity signature to confirm device orientation changes consistent with a live scan.

Sensor Fusion Check

Server compares statistical features (variance, entropy) against known live-interaction templates; flags robotic or replayed inputs.

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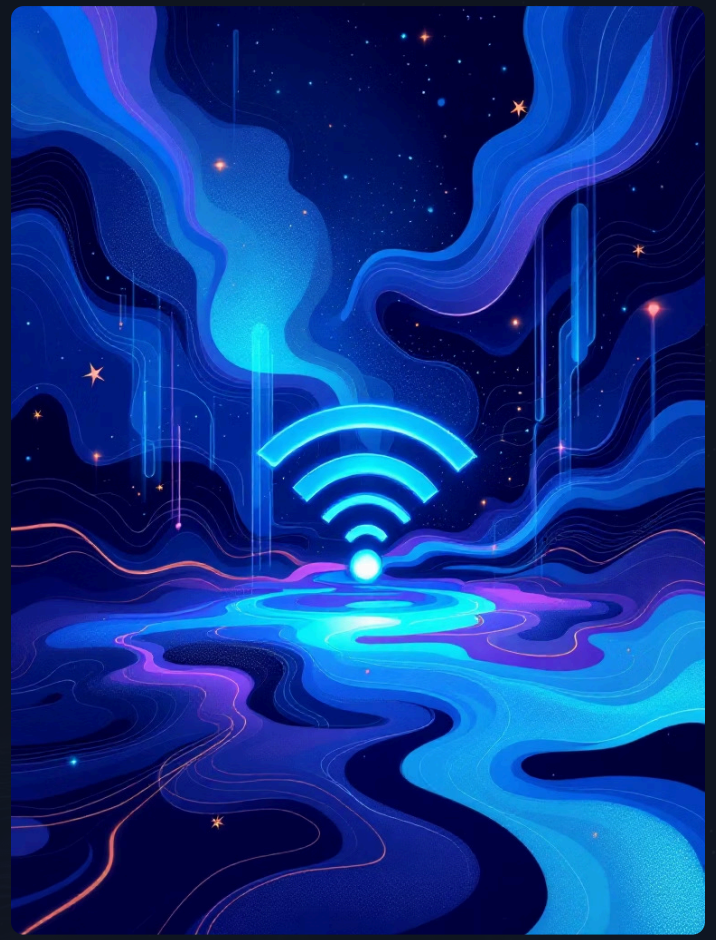

7. Optional Indoor Wi-Fi Verification

Augments GPS when inside buildings:

- Client collects nearby SSIDs + RSSI vectors (hashed locally for privacy)
- Server matches fingerprint against known classroom profiles
- Used as conditional factor when GPS accuracy < threshold
- Privacy: no raw SSIDs stored—only irreversible hashes and similarity scores

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Receive **Verify QR** **Geo-fence** **Motion** **WiFi Match**

All validations are modular microservices. Each produces a confidence score; a weighted policy engine computes final decision and logs evidence for audits.

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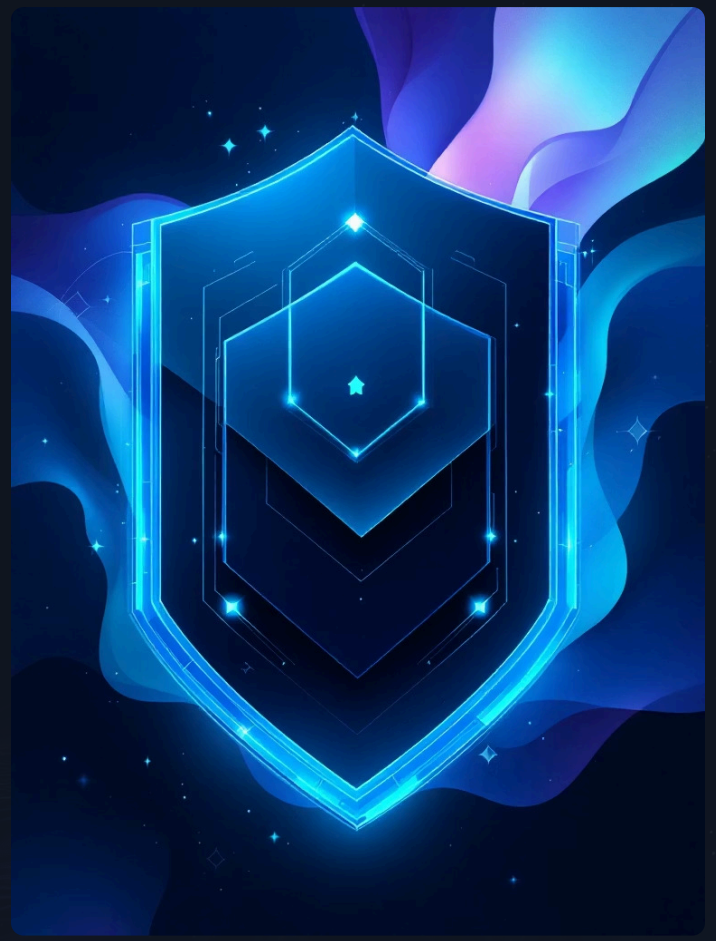

8. Security & Privacy Layers

Transport & Storage

TLS 1.3 for transit; DB encryption-at-rest. Key rotation policy enforced.

Authentication

Signed QR tokens + device attestation where available.

Integrity & Anti-Forgery

Nonces, short TTLs, sensor fusion signatures to prevent replay.

Audit & Privacy

Immutable logs, role-based access, PII minimisation (hashed fingerprints).

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